Fourier & PDEs

Saturday, 24 October 2026 1:20 pm

Solve the PDE: $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $x>0, t>0$ subject to

Initial Condition: $u(x,0) = \begin{cases} 1, & \text{if } 0<x<1. \\ 0, & \text{if } x \geq 1. \end{cases}$

Boundary Condition: $u(0,t) = 0$ for all $t>0$.

Condition at Infinity: $u \to 0$ as $x \to \infty$ and $\frac{\partial u}{\partial x} \to 0$ as $x \to \infty$.

We apply the Fourier Sine Transform (FST): $\mathcal{F}_s$ to both sides of the PDE, (See Slides 40-42.) w.r.t. x.

Recall: $\mathcal{F}_s[f(x)] = \hat{f}_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin \omega x \, dx$.

$$\mathcal{F}_s^{-1}[\hat{f}_s(\omega)] = f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f}_s(\omega) \sin \omega x \, d\omega.$$

$$\mathcal{F}_s\left[\frac{\partial u}{\partial t}\right] = \mathcal{F}_s\left[\frac{\partial^2 u}{\partial x^2}\right]$$

$$\Rightarrow \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{\partial u}{\partial t} \sin \omega x \, dx = \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{\partial^2 u}{\partial x^2} \sin \omega x \, dx.$$

$$\Rightarrow \frac{d}{dt} \sqrt{\frac{2}{\pi}} \int_0^\infty u \sin \omega x \, dx = \sqrt{\frac{2}{\pi}} \left\{ \left[\left(\frac{\partial u}{\partial x} \right)^{0} \sin \omega x \right]_0^\infty - \omega \int_0^\infty \frac{\partial u}{\partial x} \cos \omega x \, dx \right\}$$

$$= \sqrt{\frac{2}{\pi}} \left\{ 0 - \omega \int_0^\infty \frac{\partial u}{\partial x} \cos \omega x \, dx \right\}$$

$$= \sqrt{\frac{2}{\pi}} \left\{ -\omega \left[u^{0} \cos \omega x \right]_0^\infty + \omega^2 \int_0^\infty u \sin \omega x \, dx \right\}$$

$$\therefore \sqrt{\frac{2}{\pi}}\frac{d\hat{u}_s}{dt} = -\omega^2\sqrt{\frac{2}{\pi}}\int_0^\infty u\sin(\omega x)dx.$$

$$\Rightarrow \frac{d\hat{u}_s}{dt} = -\omega^2\hat{u}_s.$$

$$\therefore \frac{d\hat{u}_s}{dt} + \omega^2\hat{u}_s = 0. \quad \leftarrow \text{A variable separable ODE in } \hat{u}_s \text{ and } t.$$

$$\Rightarrow \frac{d\hat{u}_s}{\hat{u}_s} = -\omega^2 dt \quad \leftarrow \text{Separation of Variables}$$

$$\therefore \ln\hat{u}_s = -\omega^2 t + A$$

$$\Rightarrow \hat{u}_s = cAe^{-\omega^2 t} \quad \text{---- } \dagger. \text{ The solution of } *.$$

Find the Value cA – a function of ω using ICs:

$$u(x,0) = \begin{cases} 1, & \text{if } 0 < x < 1. \\ 0, & \text{if } x \geq 1. \end{cases}$$

At $t=0$: we take FST of IC:

$$\hat{u}_s(\omega,0) = \sqrt{\frac{2}{\pi}}\int_0^\infty u(x,0)\sin\omega x\,dx$$

$$= \sqrt{\frac{2}{\pi}}\left[\int_0^1 1\cdot\sin\omega x\,dx + \int_1^\infty 0\cdot\sin\omega x\,dx\right]$$

$$= \sqrt{\frac{2}{\pi}}\left[-\frac{\cos\omega x}{\omega}\right]_0^1$$

$$= \sqrt{\frac{2}{\pi}}\left[\frac{1-\cos\omega}{\omega}\right].$$

$$\hat{u}_s(\omega,t) = cA\, e^{-\omega^2 t}.$$

Put $t=0$: $cA = \hat{u}_s(\omega,0)$

$$= \sqrt{\frac{2}{\pi}}\left(\frac{1-\cos\omega}{\omega}\right).$$

$$\therefore \quad \hat{u}_s(\omega,t) = \sqrt{\frac{2}{\pi}}\left(\frac{1-\cos\omega}{\omega}\right) e^{-\omega^2 t} \quad \text{---} \ (\dagger\dagger)$$

The solution of our PDE can be found by taking the Inverse Fourier Sine Transform of $(\dagger\dagger)$:

$$u(x,t) = \mathcal{F}_s^{-1}\left[\hat{u}_s(\omega,t)\right]$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{u}_s(\omega,t) \sin\omega x \, d\omega$$

$$= \sqrt{\frac{2}{\pi}} \int_0^\infty \sqrt{\frac{2}{\pi}}\left(\frac{1-\cos\omega}{\omega}\right) e^{-\omega^2 t} \sin\omega x \, d\omega.$$

$$\therefore \quad u(x,t) = \frac{2}{\pi} \int_0^\infty \frac{(1-\cos\omega)}{\omega} e^{-\omega^2 t} \sin\omega x \, d\omega.$$

Solve the PDE: $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $x>0, t>0$ subject to

Initial Condition: $u(x,0) = \begin{cases} x, & 0 \le x \le 1. \\ 0, & x > 1. \end{cases}$

Boundary Condition: $u_x(0,t) = \frac{\partial u}{\partial x}(0,t) = 0$, for all $t>0$.

Condition at Infinity: $u \to 0$ as $x \to \infty$ and $\frac{\partial u}{\partial x} \to 0$ as $x \to \infty$.

We apply the Fourier Cosine Transform (FCT): $\mathcal{F}_c$ to both sides of the PDE, (see slides 40-42) w.r.t. x.

Recall: $\mathcal{F}_c[f(x)] = \hat{f}_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos\omega x \, dx.$

$$\mathcal{F}_c^{-1}[\hat{f}_c(\omega)] = f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f}_c(\omega) \cos\omega x \, d\omega.$$

Hence,

$$\mathcal{F}_c\left[\frac{\partial u}{\partial t}\right] = \mathcal{F}_c\left[\frac{\partial^2 u}{\partial x^2}\right].$$

$$\Rightarrow \frac{d\hat{u}_c}{dt} = -\sqrt{\frac{2}{\pi}} u_x(0,t) - \omega^2 \hat{u}_c$$

$$= -\sqrt{\frac{2}{\pi}} \times 0 - \omega^2 \hat{u}_c.$$

$$\therefore \frac{d\hat{u}_c}{dt} + \omega^2 \hat{u}_c = 0 \leftarrow \text{A Variable Separable ODE}$$

Hence, $\frac{d\hat{u}_c}{\hat{u}_c} = -\omega^2 dt \Rightarrow \hat{u}_c = A e^{-\omega^2 t}.$

Find the Value of A – a function of ω using ICs:

$$u(x,0) = \begin{cases} x, & 0 < x < 1 \\ 0, & x \geq 1. \end{cases}$$

At $t=0$: We take FCT of IC:

$$\hat{u}_c(\omega,0) = \sqrt{\frac{2}{\pi}} \int_0^\infty u(x,0) \cos\omega x \, dx.$$

$$= \sqrt{\frac{2}{\pi}} \int_0^1 x \cos\omega x \, dx$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{x \sin\omega x}{\omega} + \frac{\cos\omega x}{\omega^2} \right]_0^1$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{\sin\omega}{\omega} + \frac{\cos\omega}{\omega^2} - \frac{1}{\omega^2} \right]$$

$$\therefore \hat{u}_c(\omega,0) = \sqrt{\frac{2}{\pi}} \left[\frac{\sin\omega}{\omega} + \frac{\cos\omega}{\omega^2} - \frac{1}{\omega^2} \right]$$

But $\hat{u}_c(\omega,t) = A e^{-\omega^2 t}$. Put $t=0$

$$\therefore A = \sqrt{\frac{2}{\pi}} \left[\frac{\sin\omega}{\omega} + \frac{\cos\omega}{\omega^2} - \frac{1}{\omega^2} \right].$$

Therefore,

$$\hat{u}_c(\omega,t) = \sqrt{\frac{2}{\pi}} \left[\frac{\sin\omega}{\omega} + \frac{\cos\omega}{\omega^2} - \frac{1}{\omega^2} \right] e^{-\omega^2 t} \quad \text{—(*)}$$

The Solution of our PDE can be found by taking the Inverse Fourier Cosine Transform of (*):

$$u(x,t) = \mathcal{F}_c^{-1}[\hat{u}(\omega,t)] = \frac{2}{\pi} \int_0^\infty \left[\frac{\sin\omega}{\omega} + \frac{\cos\omega}{\omega^2} - \frac{1}{\omega^2} \right] \cos\omega x \, d\omega.$$